

Bellabeat Case Study

Executive Summary

sleep, and daily activity consistency and translate those findings into recommendations for Bellabeat's marketing strategy. Analysis of 35 users revealed substantial differences in activity behavior, including a large lower-activity segment, higher levels of sedentary time among lower-activity users, greater day-to-day activity variability, and a much higher frequency of very-low-activity days. Sustained activity was most commonly observed during the afternoon. Based on these findings, the analysis recommends that Bellabeat focus on reducing prolonged sedentary behavior, encouraging consistent daily movement, addressing very-low-activity days, and using activity patterns to guide the timing of user engagement.

Business Task & Objectives

The primary objective is to analyze smart device usage data to gain insight on how people are already using their devices. Using this information, I will develop recommendations on how the discovered trends can inform Bellabeat's marketing strategy.

- What are some trends in smart device usage?
- How do these trends apply to Bellabeat customers?
- How could these trends help influence Bellabeat's marketing strategy?

Data & Sources

The primary dataset used for this case study is the Fitbit Fitness Tracker Data available through Mobius on Kaggle. The dataset contains fitness tracker observations collected from 35 users during a two-month period from March 12, 2016 through May 12, 2016. The available data includes daily, hourly, and minute-level measures of physical activity, intensity, steps, calories, sleep, heart rate, and weight-related measurements.

The dataset provides relevant information for examining physical activity and wellness behaviors, although limitations in sample size, data availability, and collection periods must be considered when interpreting the results.

Data Limitations & Assumptions

The data used for this case study presents the following limitations:

- The Daily_Activity dataset contains 35 unique users. Although this is sufficient to identify patterns within the sample, the relatively small sample size limits how confidently the findings can be generalized to the broader population of smart-device users.
- Sleep data was available at different levels of granularity across the two collection periods. The second period included sleep_Day, which provided daily-level sleep records, while the first period contained minute-level sleep data. The daily-level sleep dataset was selected for analysis to maintain consistency with the project's daily activity analysis and simplify interpretation.
- The Sleep_Day data is only available for the second collection period.
- The Sleep_Day data is only available for 24 users, compared to 35 users for the Daily_Activity.
- The dataset does not provide demographic or other contextual information sufficient to determine whether differences in activity behavior are associated with factors such as age, occupation, or other individual characteristics.

The data used for this case study makes the following assumptions:

- The analysis assumes that the behavioral patterns observed in this sample may provide directional insight into smart-device usage behaviors, while recognizing that the sample may not be representative of the broader population.

Data Cleaning and Preparation

The dataset contained activity information at multiple levels of granularity, including daily, hourly, and second level records. The available datasets were reviewed and will be selected based on their relevance to the business questions and compatibility of their levels of aggregation.

Data Organization:

- 1) Downloaded the provided ZIP file
- 2) Extracted the files
- 3) Organized the original files into a dedicated raw data folder
- 4) Created a separate location for working/cleaned data
- 5) Backed up the project files to Google Drive

Initial Data Assessment:

I received 20+ CSV files as part of the project dataset. Rather than manually opening each file, I used Python with the pandas library to programmatically examine the files. The script generated an inventory containing each file name, number of rows, and number of columns. This allowed me to efficiently assess the structure and consistency of the dataset before beginning data preparation and analysis.

File	Rows	Columns
3.12-4.11		
dailyActivity_merged	457	15
heartrate_seconds_merged	1,154,681	3
hourlyCalories_merged	24,084	3
hourlyIntensities_merged	24,084	4
hourlySteps_merged	24,084	3
minuteCaloriesNarrow_merged	1,445,040	3
minuteIntensitiesNarrow_merged	1,445,040	3
minuteMETsNarrow_merged	1,445,040	3
minuteSleep_merged	198,559	4
minuteStepsNarrow_merged	1,445,040	3
weightLogInfo_merged	33	8
4.12-5.21		
dailyActivity_merged	940	15
dailyCalories_merged	940	3
dailyIntensities_merged	940	10
dailySteps_merged	940	3
heartrate_seconds_merged	2483658	3

hourlyCalories_merged	22099	3
hourlyIntensities_merged	22099	4
hourlySteps_merged	22099	3
minuteCaloriesNarrow_merged	13,215,580	3
minuteCaloriesWide_merged	21645	62
minuteIntensitiesNarrow_merged	1,325,580	3
minuteIntensitiesWide_merged	21645	62
minuteMETsNarrow_merged	1,325,580	3
minuteSleep_merged	188,521	4
minuteStepsNarrow_merged	1,325,580	3
minuteStepsWide_merged	21645	62
sleepDay_merged	413	5
weightLogInfo_merged	67	8

Step 1 : Data Combination

I combined the dailyActivity_merged, hourlyIntensities_merged, and weightLogInfo_merged files from each time period into files that represented the data from the entire 2-month time period and saved them to a new destination to be examined further and cleaned. Sleep_Day will be cleaned and stored in the same location.

Data Cleaning:

Using Google Sheets, I have cleaned the **Daily_Activity** file as follows:

Step 2 : Duplicate Assessment

Daily_Activity: Checked for duplicate Id + ActivityDate combinations to ensure each participant had no more than one daily activity observation.

Sleep_Day: Checked Id + SleepDay combinations. Six duplicate rows were identified across three user/date combinations. The duplicate rows were identical, so one copy of each was removed, reducing the dataset from 413 to 410 observations.

Hourly_Intensities: Checked Id + ActivityHour combinations. One hundred rows representing 50 duplicate pairs were identified. The duplicate records were identical, so 50 redundant rows were removed. No duplicate Id + ActivityHour combinations remained after the correction.

Weight_Log_Info: Checked for exact duplicate records across all fields. No duplicate observations were identified, so no records were removed.

Step 3 : Resolved overlapping source data

Daily_Activity: The source files both contained April 12, 2016. Twenty-four user/date combinations appeared in both datasets. The records from the earlier source were removed to prevent double-counting, while the later source records were retained.

Hourly_Intensities: The two source files both contained April 12, 2016. Fifty Id + ActivityHour combinations appeared in both source periods, resulting in 100 rows representing 50 duplicate pairs. The duplicate records were identical across the intensity fields. The records from the earlier source were removed, while the corresponding records from the later source were retained to prevent double-counting hourly observations.

Step 4 : Missing values

Daily_Activity: Checked all 15 columns for blank values. No missing values were identified, so no imputation or row removal was necessary.

Sleep_Day: Checked all 5 columns for blank values. No missing values were identified, so no imputation or row removal was necessary.

Hourly_Intensities: Checked all four columns for blank values. No missing values were identified, so no imputation or row removal was necessary.

Weight_Log_Info: Checked all eight columns for blank values. Ninety-six records were missing Fat measurements, while no missing values were identified in the remaining seven fields. The missing Fat values were retained because they reflect unavailable measurements in the source data and do not prevent analysis of the other weight-related fields.

Step 5 : Data types and formatting

Daily_Activity: All activity measurement fields contained numeric values. ActivityDate was recognized as a date field. No data-type conversion was required.

Sleep_day: All numeric fields were recognized as numeric values, and SleepDay was recognized as a date/time field. No data-type conversion was required.

Hourly_Intensities: All intensity measurement fields were recognized as numeric values, and ActivityHour was recognized as a date/time field. No data-type conversion was required.

Weight_Log_Info: All weight measurement fields were recognized as numeric values, and Date was recognized as a date/time field. No data-type conversion was required.

Data Validation:

Step 6 : Validity checks

Daily_Activity

- **Negative-Value Assessment:** Checked all numeric activity fields for negative values. No negative values were identified.
- **Zero-value assessment:** Zero values were identified across several activity measures. These values were retained because zero activity can represent legitimate days of inactivity or lack of recorded activity. Six records contained zero calories; these records also showed zero activity across the remaining activity measures and were therefore retained as valid observations rather than treated as errors.
- **Range validation:** Minimum and maximum values were reviewed across all numeric activity fields. No clearly impossible values were identified. High values were retained because they fall within plausible ranges for daily activity and may represent genuine high-activity observations.
- **Activity-duration validation:** Verified that the sum of very active, fairly active, lightly active, and sedentary minutes did not exceed 1,440 minutes (24 hours) for any observation. No invalid activity-duration records were identified.

Sleep_Day

- **Negative values:** No negative values were identified in TotalSleepRecords, TotalMinutesAsleep, or TotalTimeInBed.
- **Zero values:** No zero values were identified in TotalSleepRecords, TotalMinutesAsleep, or TotalTimeInBed.

- **Range validation:** TotalSleepRecords ranged from 1–3 records, TotalMinutesAsleep ranged from 58–796 minutes, and TotalTimeInBed ranged from 61–961 minutes. No clearly impossible values were identified based on these ranges.
- **Sleep duration consistency:** Compared TotalMinutesAsleep with TotalTimeInBed for each observation to ensure that reported sleep time did not exceed total time in bed. No invalid records were identified.

Hourly_Intensities

- **Negative values:** No negative values were identified in TotalIntensity or AverageIntensity.
- **Zero values:** 13,312 records contained zero values for both TotalIntensity and AverageIntensity. These values were retained because zero intensity can represent an hour with no recorded activity intensity and does not, by itself, indicate an invalid observation.
- **Intensity calculation consistency:** Compared AverageIntensity with TotalIntensity ÷ 60 for each observation. No inconsistencies were identified.

Weight_Log_Info

- **Negative values:** No negative values were identified in WeightKg, WeightPounds, BMI, or reported Fat measurements.
- **Zero values:** No zero values were identified in WeightKg, WeightPounds, reported Fat, or BMI.
- **Weight conversion consistency:** Compared WeightPounds with the corresponding WeightKg measurements using the standard kilogram-to-pound conversion. No inconsistencies were identified beyond expected rounding.

Analysis & Methodology

Analysis Approach

Following data cleaning and validation, the analysis was conducted at both the daily-observation and user levels. Daily-level analysis was used to establish overall activity patterns, while user-level aggregation was used to compare behavioral differences across participants. Google Sheets pivot tables were used to aggregate activity measures by user, and scatter plots and correlation analysis were used to examine relationships between key behavioral variables.

User-Level Activity Analysis

A user-level pivot table was created from the cleaned Daily_Activity dataset to compare average daily steps, calorie expenditure, activity intensity, and sedentary time across the 35 participants.

Daily Step & Calorie Relationship

Analytical Question: Is higher daily step count associated with higher calorie expenditure?

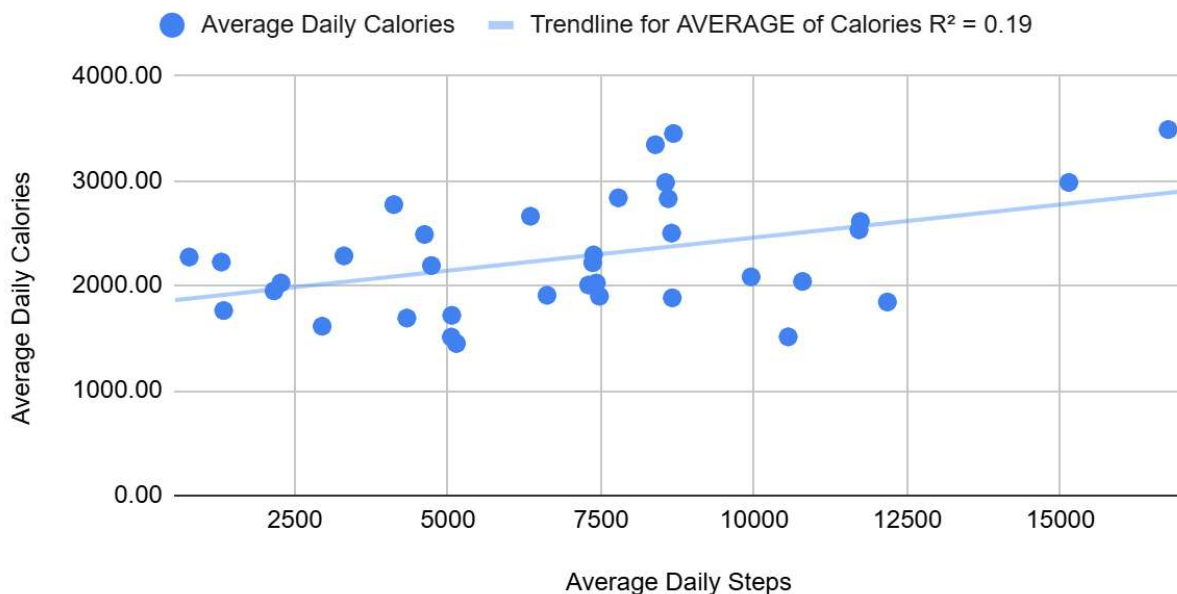
Why This Question Matters: Understanding the relationship between daily step count and calorie expenditure may help Bellabeat identify opportunities to encourage users to increase their physical activity.

Data/Analysis Used: A scatter plot was created to observe the relationship and correlation between daily steps and calorie expenditures. The value r was calculated to describe this relationship.

Visualization(s)

Daily Step and Calorie Expenditure

Average daily values across 35 users



Key Finding and Statistical Evidence: The analysis revealed a correlation coefficient of $r = 0.44$ between daily steps and calorie expenditure, indicating a moderate positive relationship between the two variables. This suggests that higher values of daily steps tend to be associated with higher values of calorie expenditure. However, the correlation does not establish a causal relationship between the variables.

Interpretation: The moderate positive relationship suggests that users who take more daily steps generally tend to expend more calories. However, the relationship is not strong, indicating that daily step count alone does not fully explain differences in calorie expenditure. Other factors, such as activity intensity, duration, and individual characteristics, may also contribute to calorie expenditure.

Business Implication: Although daily steps serve as a helpful metric for tracking movement, Bellabeat should recognize that step count alone provides an incomplete picture of calorie expenditure. Integrating step counts with additional physical activity measures could give users a more comprehensive understanding of their overall activity and wellness behaviors.

Activity & Sedentary Behavior

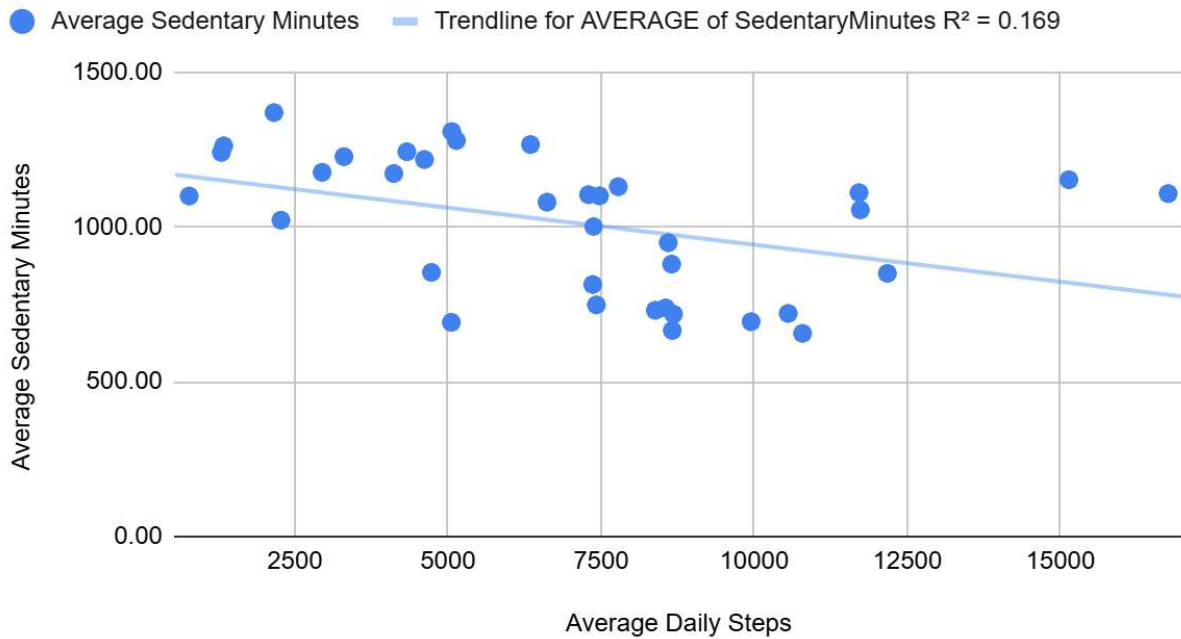
Analytical Question: Is higher daily activity associated with less sedentary time? Does spending more time in active categories correspond to less sedentary time?

Why This Question Matters: Understanding how movement and sedentary time relate may help Bellabeat identify opportunities to encourage users to become more active while reducing prolonged periods of inactivity.

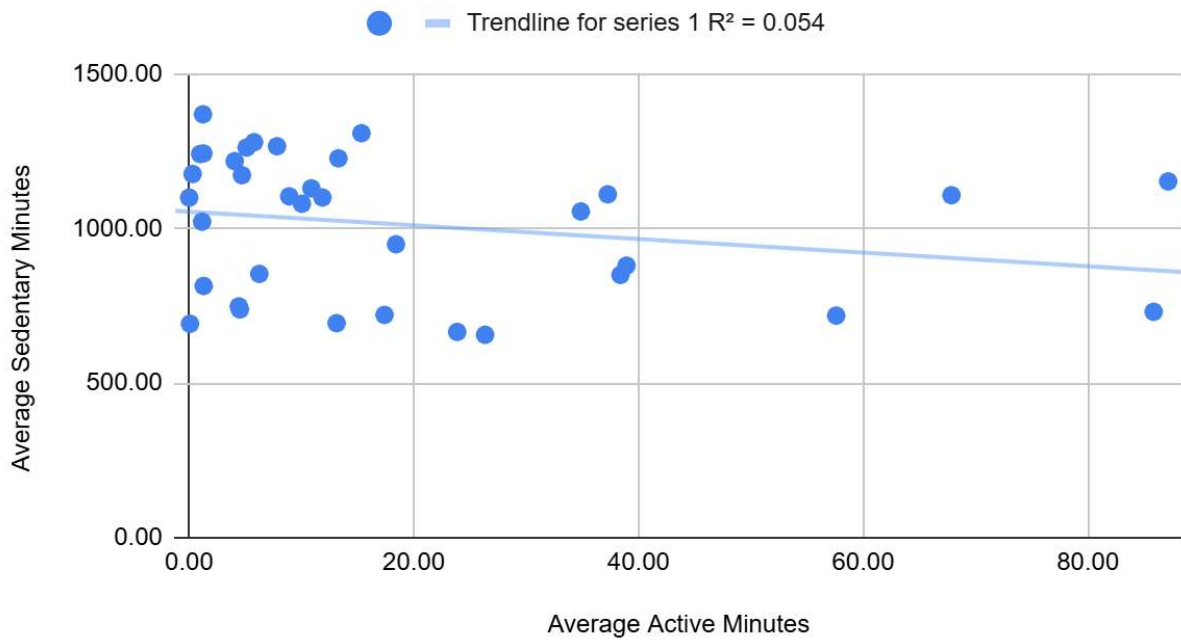
Data/Analysis Used: 2 scatter plots were created to examine the correlation between daily steps and sedentary time as well as the correlation between active time and sedentary time. The value r was calculated to describe both of these relationships.

Visualization(s):

Daily Steps and Sedentary Time



Active Time and Sedentary Time



Key Finding and Statistical Evidence: The analysis revealed a negative correlation between daily steps and sedentary time ($r = -0.41$) and a negative correlation between active time and sedentary time ($r = -0.23$). These results indicate that users with higher levels of physical activity generally tend to record less sedentary time. However, the strength of the relationships indicates that activity level alone does not fully explain differences in sedentary behavior.

Interpretation: Overall, greater physical activity is associated with lower sedentary time, but the relationships are not strong enough to suggest that increasing activity automatically eliminates prolonged periods of inactivity. Users with similar levels of activity may still have substantially different amounts of sedentary time.

Business Implication: Bellabeat should address physical activity and sedentary behavior as complementary but separate behaviors. Encouraging users to increase activity while also prompting them to break up prolonged sedentary periods could provide a more complete approach to supporting everyday wellness.

Distribution of Daily Activity

Analytical Question: How is users' daily time distributed across different activity levels, and what does this distribution suggest about overall activity and sedentary behavior?

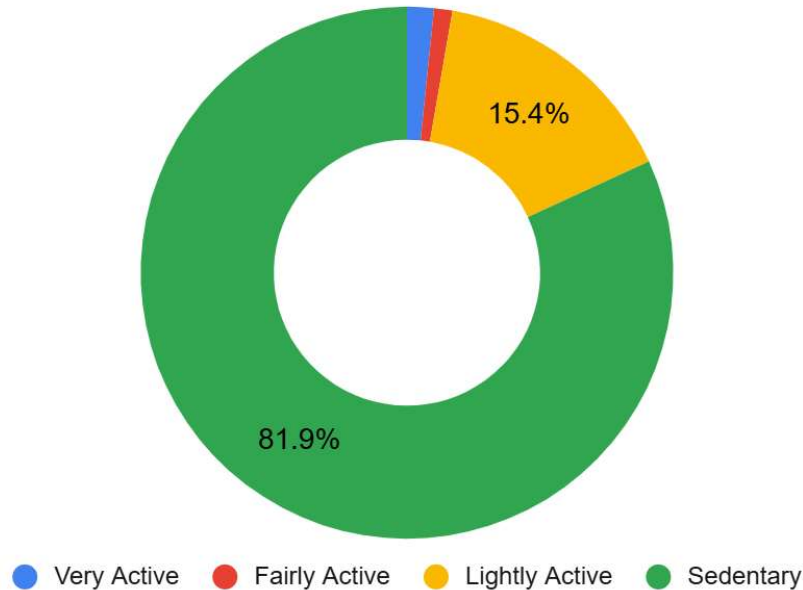
Why This Question Matters: Understanding how users typically spend their day can help Bellabeat identify behavioral patterns and opportunities to encourage greater movement throughout the day. Examining the distribution of activity levels provides context for understanding whether users' daily routines are characterized primarily by active movement or sedentary behavior.

Data/Analysis Used: A distribution chart was created to compare the average amount of time users spent in each recorded daily activity category: very active, fairly active, lightly active, and sedentary. The percentage of a 24-hour day represented by each category was also calculated to provide additional context.

Visualization(s):

Distribution of Recorded Daily Activity Time

Average across 35 users



Key Finding and Statistical Evidence: The analysis found that sedentary time represented the largest portion of users' recorded daily time, averaging 1,001.3 minutes per day, or approximately 69.5% of a 24-hour day. Lightly active time accounted for an average of 188.1 minutes (13.1%), while fairly active and very active time accounted for approximately 13.6 minutes (0.9%) and 19.9 minutes (1.4%), respectively.

Interpretation: The distribution indicates that users spent substantially more time sedentary than engaged in any level of physical activity. Although users accumulated meaningful amounts of light activity, moderate and vigorous activity represented a relatively small portion of the day. This suggests that increasing overall movement may require more than simply encouraging occasional high-intensity exercise; reducing prolonged sedentary periods and increasing light activity throughout the day may also represent meaningful opportunities.

Business Implication: Bellabeat could use these findings to promote movement as a continuous part of the user's daily routine rather than focusing exclusively on traditional exercise goals. Features such as reminders to stand, take short walking breaks, or add small amounts of movement throughout the day could help users reduce sedentary time while making activity goals feel more achievable. This approach could also create additional opportunities for users to interact with their Bellabeat devices throughout the day.

Activity Across Step Quartiles

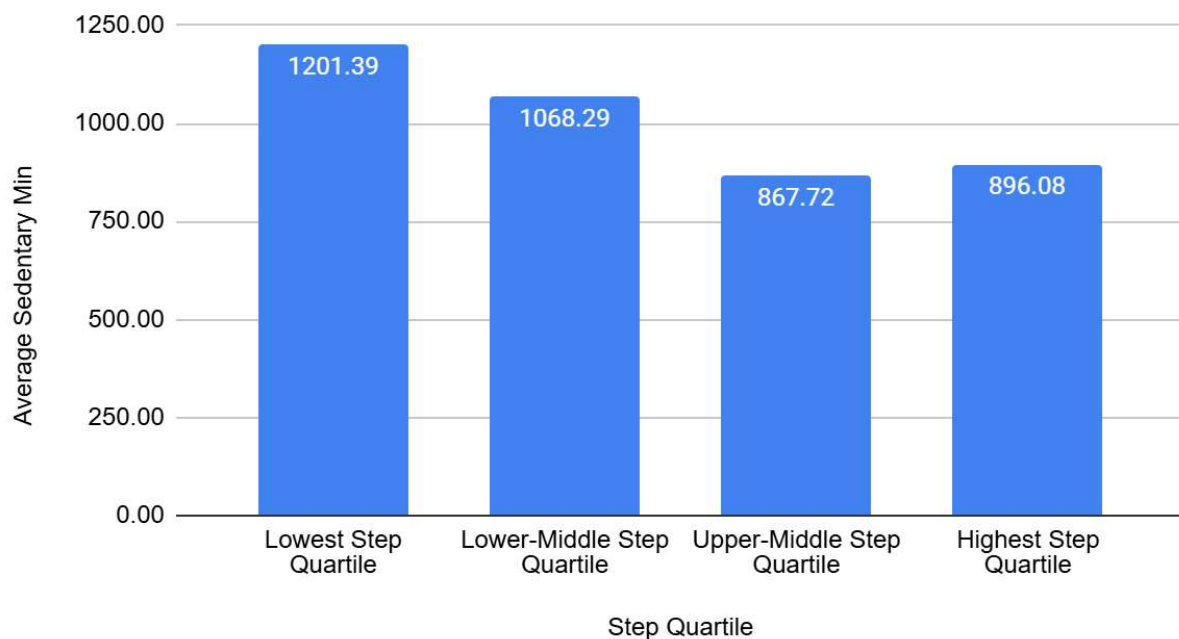
Analytical Question: How do physical activity and sedentary behavior differ across daily step quartiles?

Why This Question Matters: Examining activity patterns across step-level groups may help Bellabeat identify meaningful differences in movement and sedentary behavior between users with lower and higher daily step counts.

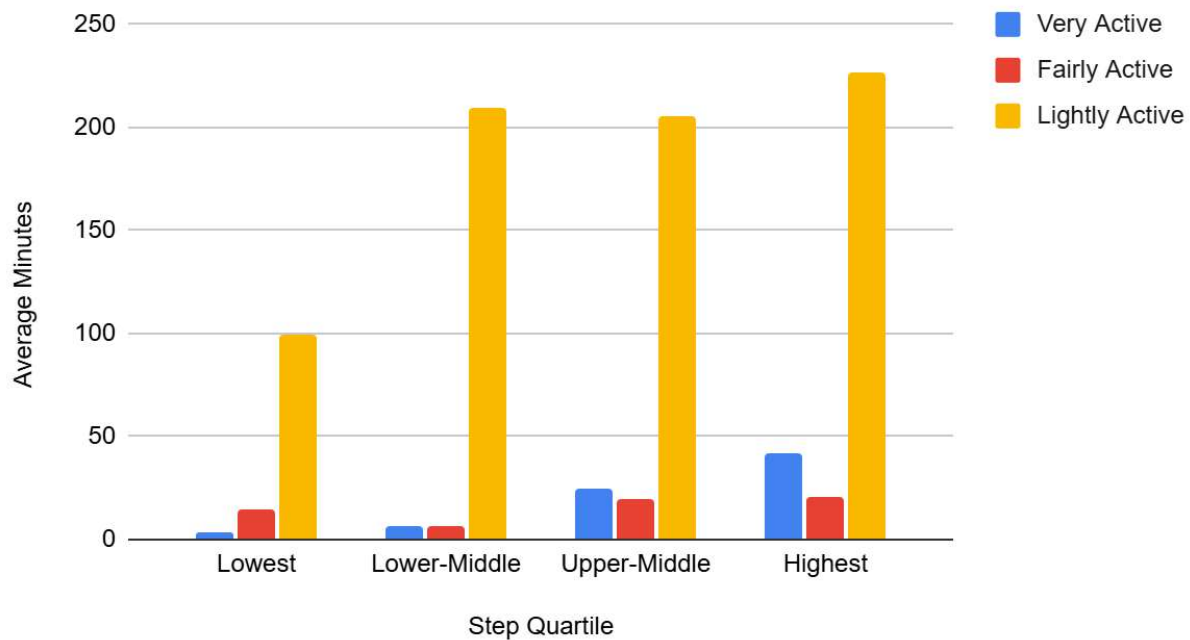
Data/Analysis Used: Users were divided into four groups based on their average daily steps, from the lowest to highest step quartile. Average active and sedentary minutes were then compared across the four quartiles to identify differences in activity patterns.

Visualization(s):

Average Sedentary Minutes by Step Quartile



Average Active Minutes by Step Quartile



Key Finding and Statistical Evidence: The analysis shows that activity levels generally increased as daily step count increased, while sedentary behavior showed the opposite pattern. Users in the highest step quartile averaged 42.2 very active minutes per day, compared with 3.1 minutes among users in the lowest step quartile. The comparison indicates substantial differences in activity levels between users with lower and higher daily step counts.

Interpretation: Users with higher daily step counts generally demonstrated a more active daily behavior pattern than users with lower step counts. In particular, the increase in very active minutes across the step quartiles suggests that higher step counts are associated with greater engagement in more intensive physical activity. At the same time, differences in sedentary time indicate that users with lower step counts may spend more of their day inactive.

Business Implication: These findings suggest that Bellabeat could use step-based activity levels to identify opportunities for personalized engagement. Rather than focusing solely on achieving a single step goal, Bellabeat could encourage users to gradually increase movement and reduce sedentary time through personalized reminders, progress goals, and activity prompts tailored to their current level of activity.

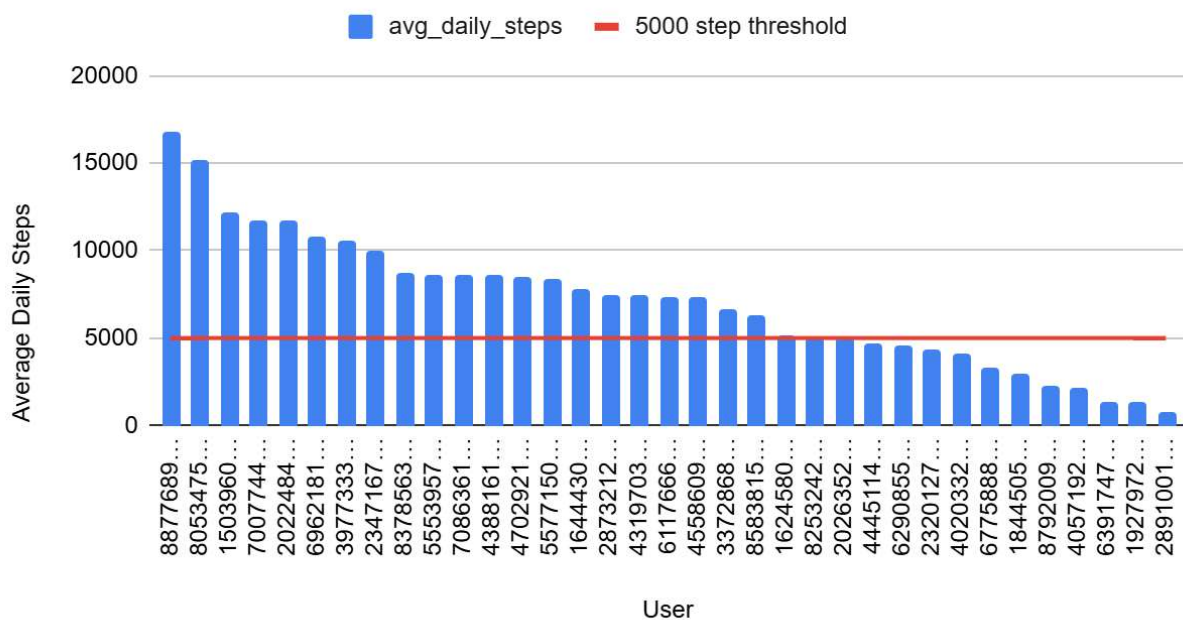
Key Findings & Visualizations

Finding 1: Activity levels vary substantially across users.

Key finding: Activity levels varied substantially across the 35 users analyzed. Average daily steps ranged from 774 to 16,759 steps per day, and 11 users (31.4%) averaged fewer than 5,000 steps per day.

Visualization:

Average Daily Steps Varied Substantially Across Users



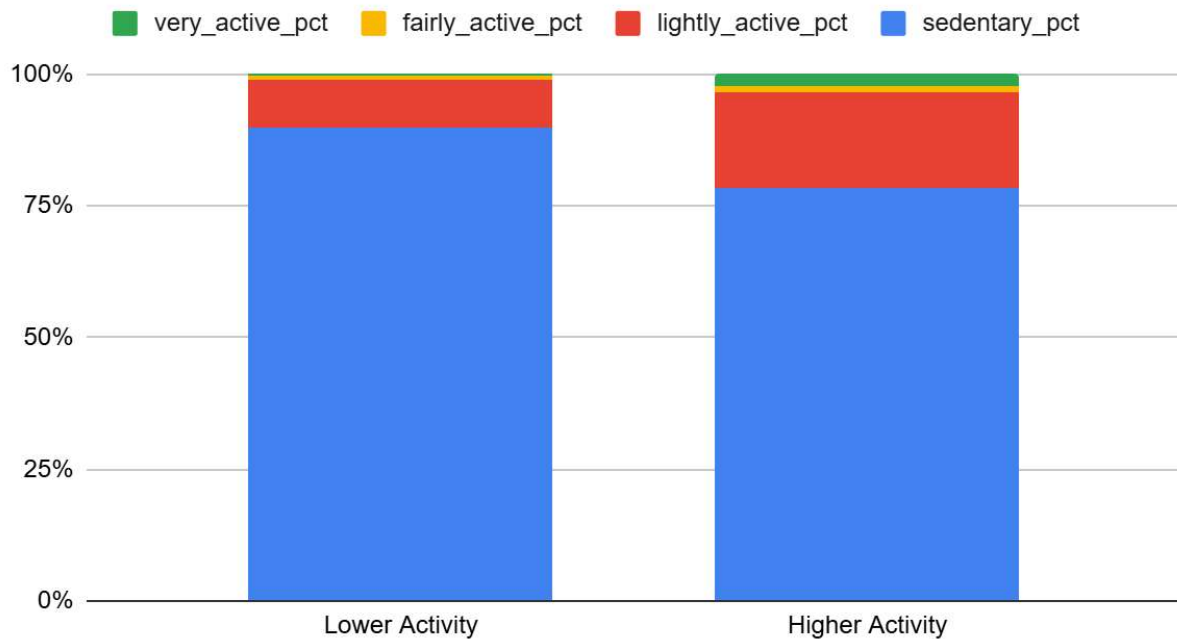
Interpretation: The wide range in average daily steps indicates substantial differences in activity behavior across users. Approximately one-third of the sample averaged fewer than 5,000 steps per day, providing a meaningful lower-activity segment for further analysis.

Finding 2: Daily activity composition differs between lower- and higher-activity users.

Key finding: Lower-activity users spent an average of 81.3% of the day sedentary, compared with 65.2% among higher-activity users. Higher-activity users also spent a greater proportion of the day in light, fairly active, and very active activity (15.0%, 1.1%, and 1.8%, respectively, vs. 8.1%, 0.8%, and 0.2%).

Visualization:

Daily Activity Composition by Activity Level



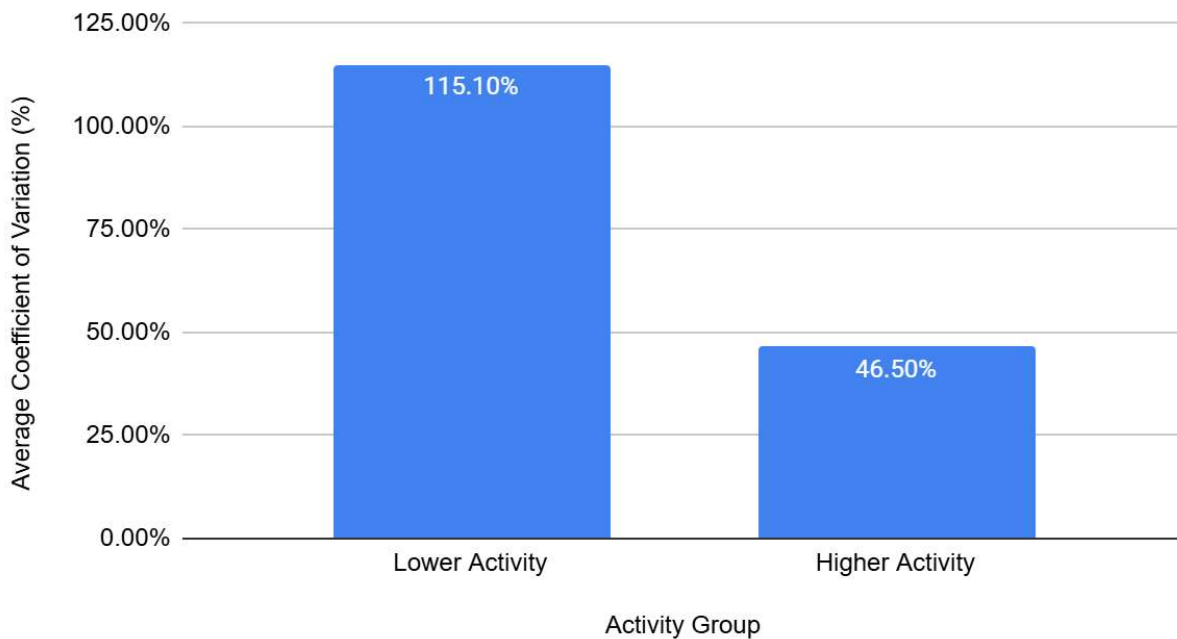
Interpretation: The difference between lower- and higher-activity users was not limited to periods of intense exercise. Higher-activity users spent substantially less time sedentary and more time engaged in light and higher-intensity activity. This suggests that increasing everyday movement, rather than focusing exclusively on intense exercise, may represent an opportunity for Bellabeat.

Finding 3: Higher-activity users maintain more consistent daily activity.

Key finding: Lower-activity users had an average coefficient of variation of 115.1%, compared with 46.5% among higher-activity users. The median comparison showed a similar pattern, with median CV values of 110.0% and 44.2%, respectively.

Visualization:

Higher-Activity Users Had More Consistent Daily Activity



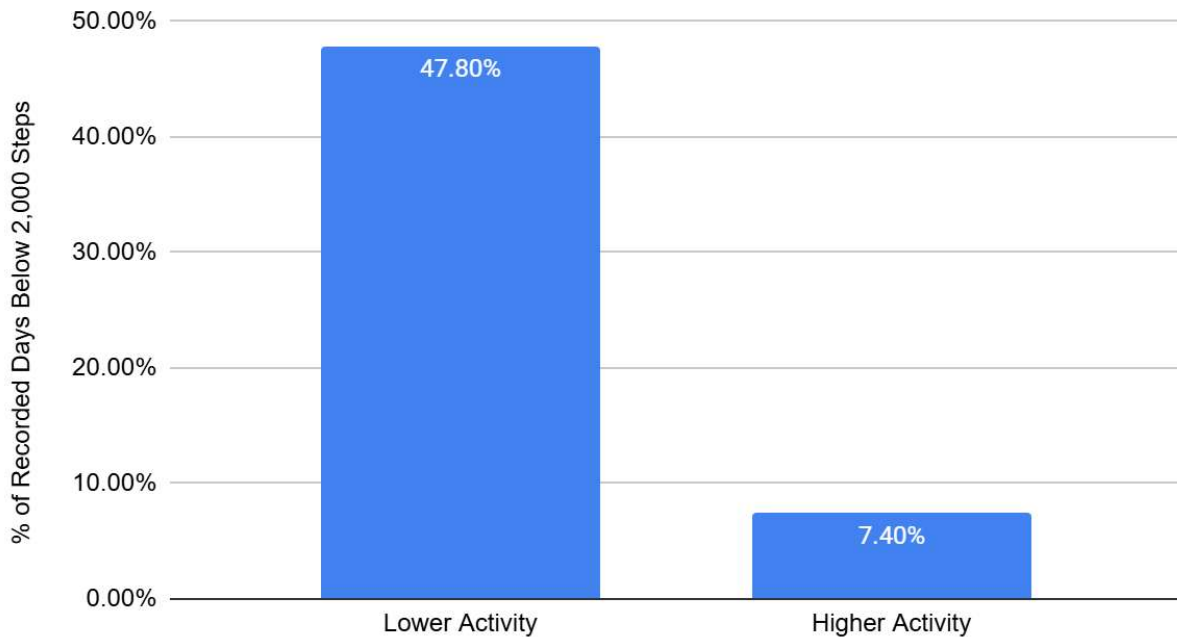
Interpretation: Higher-activity users demonstrated substantially less day-to-day variation in steps relative to their typical activity level. This suggests that higher activity in this sample was associated not only with taking more steps, but also with maintaining a more consistent daily activity pattern.

Finding 4: Lower-activity users experience very-low-activity days more frequently.

Key finding: Lower-activity users averaged 47.8% of their recorded days below 2,000 steps, compared with 7.4% among higher-activity users. The median comparison showed a similar pattern, with 54.5% of recorded days below 2,000 steps for lower-activity users compared with 4.9% for higher-activity users.

Visualization:

Lower-Activity Users Had More Very Low-Activity Days



Interpretation: Lower activity in this sample involved not only a lower average number of steps but also a much higher frequency of very-low-activity days. This suggests that helping users maintain a minimum level of daily movement may be a useful engagement opportunity for Bellabeat, rather than focusing exclusively on intense exercise.

Supporting Insight: Afternoon Activity Patterns

Among the 34 users with sufficient hourly activity data, 22 (64.7%) demonstrated their highest sustained activity during the afternoon. This pattern suggests that afternoon engagement may be a useful opportunity for Bellabeat to deliver activity prompts or challenges.

Recommendations/Action Plan

Recommendations

1. Reduce Sedentary Time

The analysis found that lower-activity users spent an average of 81.3% of their day sedentary, compared with 65.2% among higher-activity users. Bellabeat should use its wellness technology to encourage users to break up prolonged periods of inactivity. Personalized reminders could prompt users to take a short walk, stretch, or engage in another brief movement activity. These small interventions may provide a more achievable starting point for users who struggle to maintain higher levels of daily activity.

2. Address Very Low-Activity Days

Lower-activity users experienced substantially more very low-activity days than higher-activity users. On average, 47.8% of recorded days among lower-activity users included fewer than 2,000 steps, compared with only 7.4% among higher-activity users. Bellabeat could develop a feature that identifies unusually inactive days and provides personalized encouragement to increase movement. Rather than focusing only on ambitious step goals, this approach could help users gradually increase activity on their least active days.

3. Encourage Consistent Daily Movement

The analysis showed that higher-activity users were more consistent in their daily activity levels, with an average coefficient of variation of 46.5%, compared with 115.1% among lower-activity users. Bellabeat should encourage consistency by introducing achievable weekly movement goals rather than focusing exclusively on a single daily step target. For example, users could be encouraged to meet a personalized activity goal on most days of the week. This approach could help users establish sustainable habits and reduce large fluctuations in activity from day to day.

4. Leverage Afternoon Activity Patterns

The analysis found that the afternoon was the most common period for sustained activity, with 22 of 34 users (64.7%) showing their highest sustained activity during this time. Bellabeat could use this pattern to strategically time notifications, activity challenges, and movement suggestions during the

afternoon. Aligning engagement with periods when users are already more likely to be active may help reinforce existing behaviors and encourage additional movement.

Action Plan

Bellabeat should prioritize reducing sedentary behavior, addressing very low-activity days, and building consistency in daily movement. These initiatives could be incorporated into the Bellabeat app through personalized inactivity reminders, weekly activity goals, and targeted afternoon engagement. Bellabeat should then monitor user engagement and changes in activity patterns to evaluate which strategies are most effective. Because this analysis is based on a relatively small sample of Fitbit users, these recommendations should be treated as opportunities for further testing rather than definitive conclusions about all Bellabeat customers.